

Forecasting LA Home Values

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Executive summary

LA home values rose 5.6% a year from 2000 to 2026, then stalled. In August 2026 the typical LA-metro home was worth \$952,601, 1.3% below its January 2025 peak. We asked three questions: which forecasting method predicts LA home values best, do mortgage rates and local jobs improve the forecast, and can a chat LLM forecast as well as a statistical model?

Data. The Zillow Home Value Index (ZHVI) for the LA metro, January 2000 to August 2026 (320 months), plus 499 LA-metro neighborhoods, the 30-year fixed mortgage rate from FRED, and monthly payroll jobs for LA and Orange County from the Bureau of Labor Statistics.

Method. We compared 14 models, from a naive drift baseline to LSTM and GRU networks and Amazon's Chronos foundation model. Four use outside information: ARIMAX with mortgage rates, ARIMAX with rates and payroll jobs, a VAR, and a dynamic factor model built from 405 neighborhoods. We also asked three chat LLMs (Claude Opus 5.5, GPT-5.5 and Gemini 3.1 Pro) to forecast at the same dates. Every model was scored two ways: a blind 80/20 test and a rolling-origin test (18 forecast origins, 1 to 12 months ahead). We then applied the course's diagnostic tools: Granger causality and impulse responses, cross-section and panel regressions, structural-break tests and a training-window study.

Findings

1. ARIMAX with the mortgage rate lagged 24 months had the lowest rolling error: 2.67% on average over 1 to 12 months, slightly ahead of ARIMA at 2.77%.
2. At 12 months ahead, no model beat a naive drift (4.92% error vs 4.98% for ARIMAX).
3. Machine learning, deep learning and the Chronos foundation model did not beat simple models. LightGBM won the single blind test (2.2%) but placed tenth of 14 in the rolling test; Chronos placed ninth.

4. Chat LLMs only looked better than our models when they could tell which years they were forecasting. Told the place and dates, Claude, GPT and Gemini averaged 1.4% to 1.6% error, far ahead of every statistical model. With place, dates and rates hidden, Claude and GPT matched ARIMAX (2.67% and 2.74%) and Gemini did worse (3.00%). All 18 test dates fall inside the models' training data, so the first numbers mostly measure memory.
5. Mortgage rates move LA prices slowly. In a VAR, a 1-point rate rise lowers home values about 4.5% after one year and 9.2% after two. Rates Granger-cause prices ($p = 0.05$), not the reverse ($p = 0.51$). Local payroll jobs add nothing: ARIMAX with jobs placed third (2.81%), and the jobs signal vanishes once the 2020-21 COVID job swings are left out.
6. The 2022 rate shock hit expensive LA County neighborhoods hardest ($t = -7.7$), a pattern that reverses in Orange County. Over the last year the coastal high end led again, while central LA and fire-hit Pacific Palisades fell.
7. Long history and local models win. Training on all data since 2000 beat starting in 2008, 2012 or 2016, and separate ARIMA models per neighborhood beat one shared factor (2.8% vs 4.1% error).

Outlook. Our models point to a 4.6% to 4.8% rise for the LA metro by August 2027, to about \$998,000. The 80% interval is wide, from -6% to +17%. Forecasts for individual neighborhoods range from -2.1% to +12.0%.

Purpose and motivation

LA is one of the most expensive housing markets in the US, and a home is the largest asset most households own. Between 2021 and 2023 the 30-year mortgage rate jumped from 2.7% to 7.6% (monthly averages). That makes LA a good test case: did a shock that large make home values harder to forecast, and do rates themselves help predict where prices go?

A good forecast matters to several groups. Buyers and sellers time transactions around it. Lenders and investors price risk with it. City and county planners use it to project property-tax revenue and affordability.

The project has three goals:

1. **Forecast** LA metro home values 12 months ahead, with an honest error band.
2. **Compare methods.** Test whether machine learning, deep learning and chat LLMs beat classical time-series models on a monthly housing index, under a fair evaluation.

3. **Explain the shock.** Measure how much mortgage rates and local jobs add to the forecast, and which LA neighborhoods the 2022 rate shock hit hardest.

Data

All series are monthly, public, and run from January 2000 to August 2026.

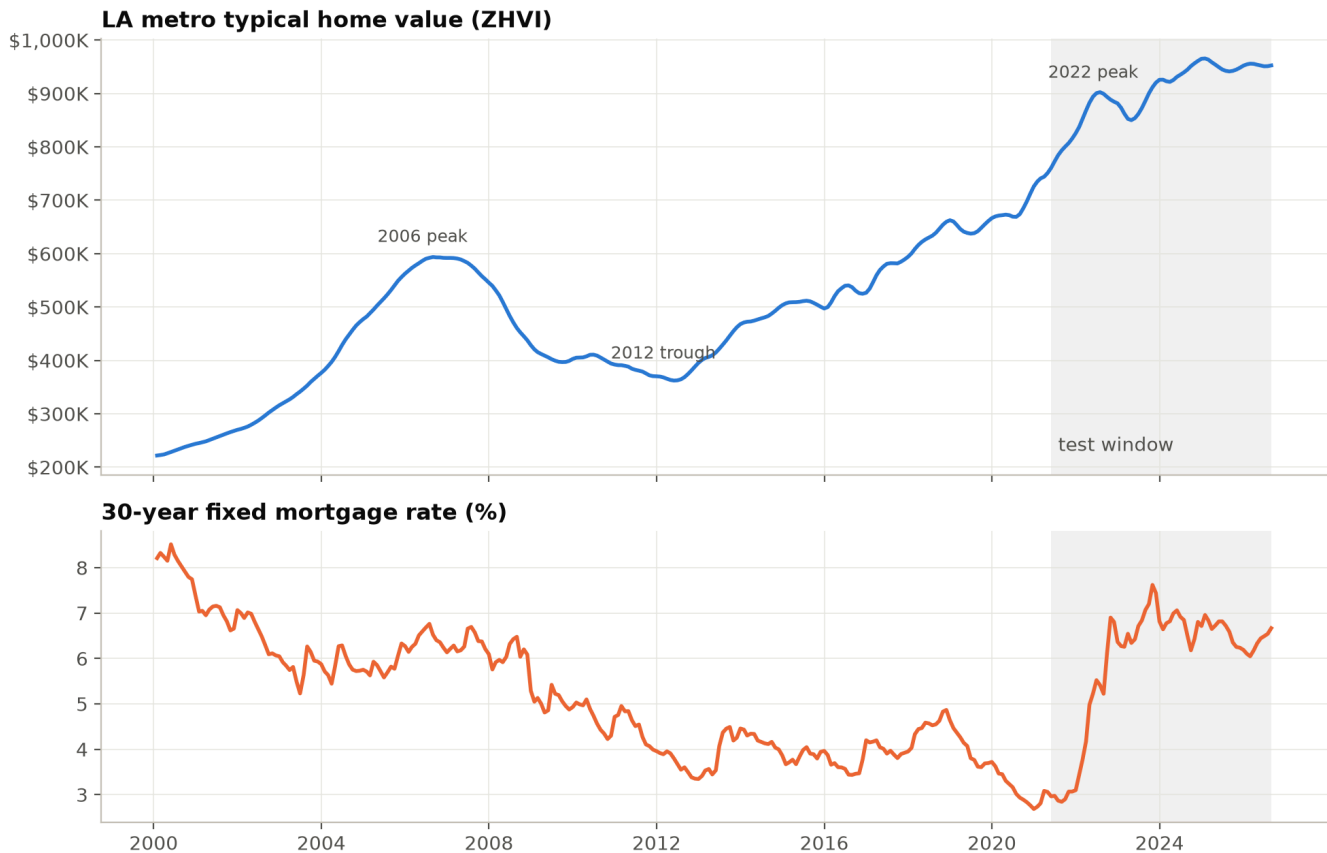
Series	Source	Coverage	Role
ZHVI, LA metro (Los Angeles–Long Beach–Anaheim)	Zillow Research	320 months	Forecast target
ZHVI, LA-metro neighborhoods	Zillow Research	499 neighborhoods (361 LA County, 138 Orange County); 405 with full 2000–2026 history	Neighborhood analysis, factor model
30-year fixed mortgage rate (MORTGAGE30US)	FRED, Freddie Mac	Weekly, averaged to monthly	Explanatory variable
Payroll jobs, total nonfarm, LA County and Orange County (SMU063108400000000001, SMU061124400000000001)	BLS, via FRED	Monthly, not seasonally adjusted (we use 12-month growth)	Explanatory variable

ZHVI measures the typical value of homes in the 35th to 65th percentile range, covering single-family homes and condos. We use Zillow's smoothed, seasonally adjusted version. That choice has two consequences we come back to later: the series has almost no seasonality left, and month-to-month changes are very persistent.

Neighborhoods use Zillow's own neighborhood boundaries. The shock and trend analysis uses the 496 neighborhoods with data since 2020; forecasting uses the 405 with complete history. Zillow's full neighborhood file is 103 MB, so we ship only the LA subset (2.7 MB) and link the source. Map positions are neighborhood centroids geocoded from OpenStreetMap.

Exploratory analysis

LA home values are not a steady upward line. They went through one full boom and bust, a long recovery, a pandemic surge and a stall once rates jumped.



- **Long-run growth:** 5.6% a year from January 2000 to August 2026.
- **2006 to 2012 bust:** values fell 37.2% from the May 2006 peak to the February 2012 trough, and did not regain the 2006 peak until December 2017.
- **2020 to 2022 boom:** with mortgage rates near 3%, values rose 34% from February 2020 to the 2022 peak.
- **Since 2022:** rates more than doubled. Values dipped in 2023, recovered to a new peak of \$965,634 in January 2025, and have drifted sideways since.

Stationarity

Log home values clearly have a unit root (ADF $p = 0.56$), so every model works with differences or growth rates. Monthly growth is borderline (ADF $p = 0.16$). Because Zillow smooths the index, monthly growth is highly persistent: its autocorrelation is 0.95 at one

month and still about 0.5 at twelve. That persistence means a model can forecast next month almost perfectly just by repeating this month, so a 1-month error tells us little. We judge models on 6- and 12-month horizons instead.

Do mortgage rates lead home values?

In the training period (2000 to April 2021), the 12-month change in mortgage rates lines up with LA home-value growth over the following two years. Its correlation with year-over-year home-value growth is +0.33 in the same month, +0.11 six months later, then -0.14, -0.29 and -0.35 after 12, 18 and 24 months.

In the same month the correlation is positive, because rates and prices both rise in a strong economy. The negative effect shows up 12 to 24 months later. That delay shapes the ARIMAX model in the next section.

Models and why we chose them

We picked 14 models to cover the course's four families: simple baselines, classical time series, machine and deep learning (including a foundation model), and models that bring in outside information. Every model forecasts log home values. The machine-learning and neural models predict monthly growth and roll forward month by month, because trees cannot extrapolate a price level they have never seen.

Model	Family	Inputs	Why it is here
Naive drift	Baseline	LA's own average monthly growth	The bar every other model must clear
ARIMA(1,1,2) with drift	Classical	LA history only	Order chosen by AIC (<code>auto_arma</code>) on the training set
Holt, damped trend	Classical	LA history only	Exponential smoothing. No seasonal term, since the series is already seasonally adjusted
Prophet	Classical	LA history only	Piecewise trend with automatic changepoints

Model	Family	Inputs	Why it is here
Prophet + COVID & rates	Classical	LA history + mortgage rate	The Assignment 2 fix: the 2020-22 boom as a "holiday" and the rate (lagged 24 months) as a regressor
XGBoost, LightGBM	Machine learning	Growth lags 1 to 12, 3/6/12-month moving averages	Can capture nonlinear momentum
LSTM, GRU	Deep learning	Previous 24 months of growth	Learn sequence patterns. 32 hidden units, early stopping on the last 10% of training data
Chronos	Foundation model	LA history only	Amazon's pretrained <code>chronos-bolt-base</code> , used zero-shot: never trained on our data
ARIMAX	Outside information	LA history + mortgage rates	Tests whether rates add forecasting power
ARIMAX + jobs	Outside information	LA history + mortgage rates + payroll job growth	Tests whether local jobs add to rates
VAR	Outside information	LA growth + monthly rate changes, 11 lags	Forecasts rates and home values together, so it needs no assumed rate path
Factor VAR	Outside information	LA growth + 2 factors from 405 neighborhoods	Big-data idea: let many neighborhoods inform the metro forecast

ARIMAX design

The exploratory analysis showed rates affect home values with a long delay, so ARIMAX models monthly growth as ARMA(1,2) errors plus the 12-month change in the mortgage rate lagged L months. We tried $L = 12, 15, 18, 21$ and 24 on the training data, and AIC picked 24.

The estimated coefficient is -0.063 (standard error 0.045 , $p = 0.16$). A 1-point rise in rates over a year lowers monthly growth by about 0.06 points two years later, roughly 0.75% of home value over the following year. The 12-month rate change peaked at 3.8 points in October 2022, which implies a drag of roughly 3% on home values spread over 2024. The effect has the expected sign and size, but it is not statistically significant, and adding the rate improves AIC only slightly. It is weak evidence, not proof.

A 24-month lag has a practical advantage: a 12-month forecast uses only rates we have already observed, so ARIMAX needs no guess about future rates. In the 64-month blind test it does eventually need future rates. There we show two versions: one that holds the last rate flat (a genuine forecast) and one that uses the rates that actually happened (a conditional benchmark, not a real forecast).

Dynamic factor model

The 405 neighborhoods with full history move together. One principal component of neighborhood growth explains 83.5% of their variance in the training period and tracks metro growth almost exactly (correlation 0.99). The second and third factors add 4.6% and 2.3%. The factor VAR uses the first two factors, re-estimated at every forecast origin on data up to that date, so no future information leaks in.

Evaluation design

We score every model out of sample in two ways, because a single train/test split can reward luck.

1. **Blind 80/20 test.** Train on January 2000 to April 2021 (256 months) and forecast May 2021 to August 2026 (64 months) in one shot, with no updates. This window contains the whole rate shock, which makes it a hard test.
2. **Rolling-origin test.** Starting at April 2021 and moving forward 3 months at a time, refit every model on all data up to that point and forecast the next 12 months. That gives 18 origins, from April 2021 to July 2025. Every model sees exactly the same information at every origin.

Model orders (ARIMA, ARIMAX lag, VAR lags) are chosen once on the training set. At each rolling origin only the parameters are re-estimated.

Metrics. In the blind test we report MAE and RMSE in dollars, MAPE in percent, and directional accuracy (the share of months where the forecast gets the direction of change right). In the rolling test we report mean absolute percentage error at 1, 3, 6 and 12

months ahead, and the average over all 12 horizons. That average is our main ranking metric.

Results: LA metro

ARIMAX and ARIMA forecast best in the rolling test, but the margins are small, and a naive drift is just as good 12 months out.

Rolling-origin test (main result)

Mean absolute % error across 18 forecast origins, sorted by the 1 to 12 month average:

Model	1 month	3 months	6 months	12 months	Average, 1-12
ARIMAX	0.09	0.80	2.45	4.98	2.67
ARIMA	0.09	0.78	2.53	5.37	2.77
ARIMAX + jobs	0.12	0.94	2.63	5.12	2.81
VAR	0.15	0.94	2.56	5.74	2.93
LSTM	0.13	0.89	2.95	5.58	3.02
GRU	0.11	0.93	2.99	5.86	3.18
Naive drift	0.69	1.93	3.49	4.92	3.27
Holt, damped	0.29	1.15	3.04	6.34	3.39
Chronos	0.69	1.61	3.44	6.13	3.56
LightGBM	0.22	1.23	3.32	6.97	3.59
Factor VAR (neighborhoods)	0.16	1.15	3.56	6.60	3.64
XGBoost	0.25	1.25	3.40	6.82	3.65
Prophet + COVID & rates	3.27	4.08	5.32	6.45	5.12
Prophet	5.00	5.77	6.73	8.62	6.89

What the table shows:

- **Rates help, but only at longer horizons.** ARIMAX and ARIMA are tied through 3 months. By 12 months ARIMAX is 0.4 points better (4.98% vs 5.37%), which is where a 24-month rate lag should matter.
- **Local jobs don't help.** ARIMAX + jobs places third (2.81%), behind ARIMAX with rates alone (2.67%), and is a little worse at every horizon.

- **Nothing beats naive drift at 12 months.** A model that simply extends LA's long-run growth is as accurate a year out as anything we built. The fitted models earn their keep at 1 to 6 months.
- **Machine learning, deep learning and the foundation model don't win.** LSTM and GRU land mid-table, Chronos ninth and the tree models tenth and twelfth. With about 250 monthly observations there isn't much nonlinear structure to learn.
- **Prophet is off from the first month.** It fits a smooth trend curve and doesn't anchor to the latest value, so it starts several percent off. The Assignment 2 fix cut its error by a quarter (6.89% to 5.12%) but not enough to catch up.

Blind 80/20 test

Model	MAE (\$)	RMSE (\$)	MAPE (%)	Direction right (%)
LightGBM	20,365	25,872	2.2	55
ARIMAX (realized rates)*	25,845	33,834	2.8	66
GRU	27,185	36,994	3.0	66
ARIMAX + jobs	32,900	38,662	3.6	66
ARIMAX	34,948	46,536	3.8	66
Holt, damped	37,033	41,154	4.0	66
Naive drift	40,259	46,423	4.5	66
Prophet + COVID & rates	47,108	53,517	5.3	63
Prophet	49,076	55,419	5.5	64
ARIMA	58,112	74,679	6.2	66
VAR	110,794	134,068	11.9	66
LSTM	126,961	153,778	13.6	66
Chronos	142,997	162,980	15.4	53
XGBoost	144,235	175,061	15.5	66
Factor VAR (neighborhoods)	197,913	231,503	21.3	66

*Uses the mortgage rates that actually happened, so it is a benchmark, not a real forecast. Direction right is close to useless here: LA values rose in 42 of 64 test months, so always predicting "up" scores 66%.

The blind test ranks the models very differently, for three reasons:

- **Momentum models overshoot.** The training period ends in April 2021, in the middle of the pandemic boom. VAR, LSTM and XGBoost extend that boom, miss the 2022 stall, and end the test window roughly 25% to 35% too high; the neighborhood factor VAR overshoots even further.
- **LightGBM's win is mostly luck.** It is the only model whose forecast bends, and it happened to bend in the right years. In the rolling test, where it gets 18 tries instead of one, it ranks tenth.
- **Rates explain part of the stall.** ARIMAX with the realized rates (2.8%) beats ARIMAX holding rates flat (3.8%). Knowing the rate path would have helped, but nobody knew it in April 2021.

Chronos made the opposite mistake: it was the only model to forecast a sustained decline, and it ended the window about 26% too low.

Can a chat LLM forecast? Claude, GPT and Gemini

Chat LLMs beat every statistical model, but only when they could tell which years they were forecasting. With those clues removed, the best of them matched ARIMAX and none beat it.

We gave three LLMs the same task as our models. At each of the 18 rolling origins, each LLM saw the monthly series up to that date and was asked for the next 12 months as a list of numbers, with no code or tools allowed. We used Claude Opus 5.5 (through the Claude Code command line, with its tools turned off), GPT-5.5 and Gemini 3.1 Pro (through their APIs): 162 prompts in all, each run once and scored exactly like the models above.

The risk is look-ahead. All 18 origins, April 2021 to July 2025, fall before these models' training cutoffs, so an LLM may have read what LA prices did next. We tested for it with three versions of every prompt:

Prompt	What the LLM sees
Named	"Los Angeles metro, Zillow ZHVI", real dates and dollar values, plus the mortgage rate

Prompt	What the LLM sees
Blind	No place or dates, values rebased to 100, the real mortgage rate
Masked	Only the last 10 years of values, rebased so the latest month = 100; no rate, no dates

An LLM that forecasts from the numbers alone should score about the same on all three.



They don't. With named or blind prompts the three LLMs averaged 1.4% to 2.1% error, better than ARIMAX (2.67%), and their 12-month errors (1.9% to 2.9%) were far below every model's (4.9% or more). With masked prompts Claude scored 2.67%, the same as ARIMAX, GPT 2.74% and Gemini 3.00%. Their 12-month errors rose to 4.5% to 4.8%, about the level of naive drift (4.92%).

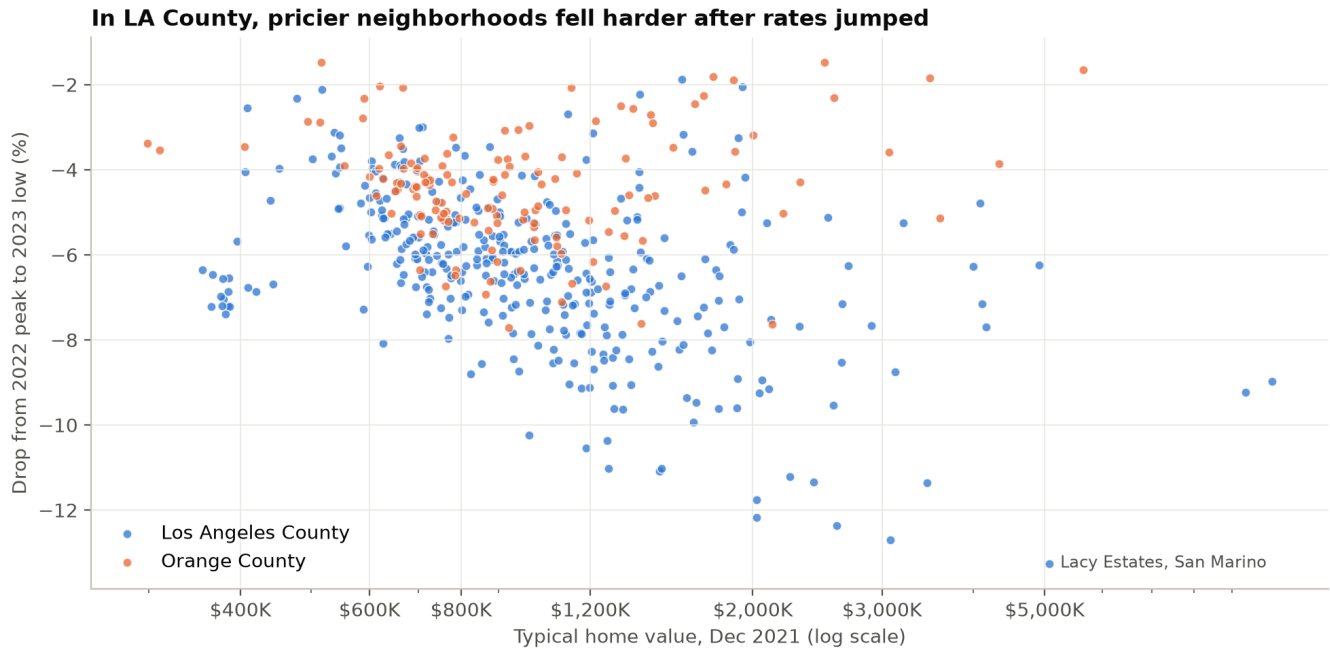
The blind prompt still gives the dates away. The mortgage rate's jump from 3% to 7% in 2022 is unique, so an LLM that recognizes it can place the series in time and recall what followed. Once the period is hidden, the best LLMs forecast about as well as ARIMAX. That is respectable for a model that fits no parameters, but it is not better, and a naive backtest would have told us the opposite. An LLM can only be backtested fairly on dates after its training cutoff.

Neighborhoods: what the metro average hides

The metro average hides a split market. Four years after rates jumped, 152 of 496 neighborhoods are still below their 2022 peak, while several Irvine neighborhoods are more than 35% above theirs.

The 2022 rate shock, neighborhood by neighborhood

For each neighborhood with data since 2020 we measured the drop from its 2022 peak to its low point between mid-2022 and 2023, and where it stands today compared with that peak.



In LA County, the more expensive the neighborhood, the harder it fell (correlation between log price and the drop: -0.45). The five biggest drops were all in San Marino, led by Lacy Estates (-13.3%). A buyer of a \$2 million home needs a much larger loan, so a jump from 3% to 7% hits that buyer harder in dollars. In Orange County there is no such pattern (correlation $+0.10$).

	LA County	Orange County
Boom, Feb 2020 to 2022 peak	+32.5%	+37.9%
Drop, 2022 peak to 2023 low	-6.3%	-4.3%
August 2026 vs 2022 peak	+1.2%	+15.3%

Medians across neighborhoods.

The neighborhoods furthest below their 2022 peak are central and urban: SoFo in West Hollywood (-30.4%), MacArthur Park (-24.0%), Arlington Heights (-19.4%), Hollywood (-17.9%) and Downtown Santa Monica (-17.7%). These areas have a high share of condos.

One hypothesis is that condos recovered more slowly than single-family homes; Zillow's separate condo index could test it.

Which neighborhoods are rising and falling now

Over the last 12 months (August 2025 to August 2026), 220 neighborhoods rose more than 2%, 216 were flat and 60 fell more than 2%.

	Rising	Flat	Falling
LA County	133	174	51
Orange County	87	42	9
City of Los Angeles (part of LA County)	22	56	18

The pattern has flipped since 2023. The high end that fell hardest is now leading: Beverly Hills Gateway (+17.6%), The Flats in Beverly Hills (+16.8%), North of Montana in Santa Monica (+15.9%) and coastal Orange County (Surfside +14.6%, Huntington Harbour +13.4%). The weakest neighborhoods are central LA (MacArthur Park -8.4%, Westlake -5.2%, Koreatown -5.0%, Downtown -4.9%), West Hollywood North (-7.4%) and Pacific Palisades (-7.3%), where the January 2025 fire, not the market, drives the decline. Only 22 of the 96 City of LA neighborhoods are rising.

An interactive map of every neighborhood, with its value, trend, 2022 drop, forecast and price history, is online at neiltd.github.io/la-housing-forecast.

Forecasting every neighborhood: shared factor or separate models?

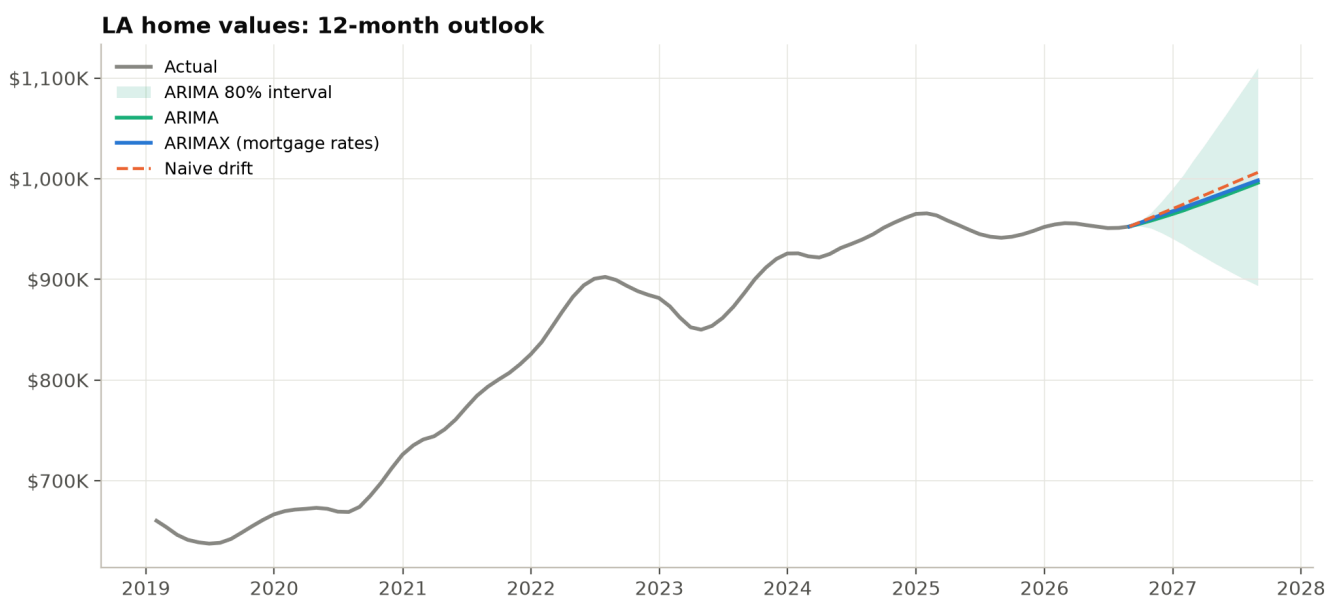
We forecast all 405 neighborhoods with full history 12 months ahead from 5 yearly origins (April 2021 to April 2025) with three methods:

Method	Average error, 1-12 months	Error at 12 months	Best for (% of cases)
ARIMA per neighborhood	2.84%	5.87%	41%
Naive drift per neighborhood	3.53%	6.89%	28%
Factor model (2 shared factors)	4.08%	8.20%	31%

Separate ARIMA models win. The factor model loses even though one factor explains 84% of neighborhood movement. The other 16% is each neighborhood's own trend, like the central-LA slump or the Irvine climb, and a two-factor model throws it away. Over 12 months those local trends add up to most of the forecast error. For the metro series the factor VAR also failed to beat plain ARIMA (3.64% vs 2.77%), because a factor that tracks the metro almost exactly carries no new information about it.

Outlook: September 2026 to August 2027

Our best model, ARIMAX, forecasts the typical LA-metro home at \$998,391 in August 2027, up 4.8% from \$952,601 today. The forecasts agree closely, but the uncertainty is wide.



Model	August 2027 forecast	Change from August 2026
ARIMAX (mortgage rates)	\$998,391	+4.8%
ARIMA	\$996,023	+4.6%
Naive drift	\$1,006,266	+5.6%
ARIMA 80% interval	\$893,893 to \$1,109,822	−6.2% to +16.5%

Because of the 24-month lag, the next 12 months of the ARIMAX forecast depend on rate changes from September 2023 to August 2025, all already observed. Rates set in 2026 will first show up in 2028. The interval is the honest headline: the model gives a 10% chance of a fall of more than 6% and a 10% chance of a rise of more than 16%.

By neighborhood

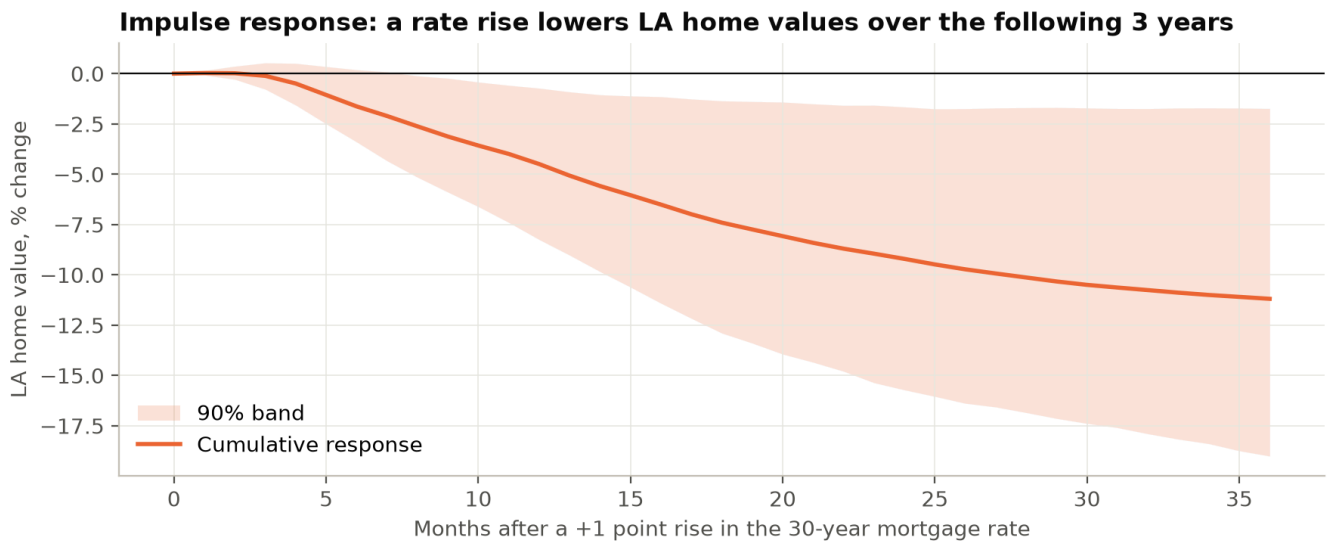
The median neighborhood forecast is +5.1%, and 80% of neighborhoods fall between +2.1% and +7.9%. Orange County neighborhoods are forecast to rise more (median +6.5%) than LA County neighborhoods (median +4.8%). Ten of 405 are forecast to fall, led by East Agoura in Agoura Hills (−2.0%). These forecasts mostly extend each neighborhood's recent trend, so they cannot see one-off events: Pacific Palisades' +5.7% forecast, for example, is a mechanical rebound after the fire, not a market signal.

Applying the course toolkit

Beyond the forecasting race, we used tools from the lectures and assignments to answer four specific questions.

Do mortgage rates drive LA prices? Granger causality and impulse response (slides 6, Assignment 4)

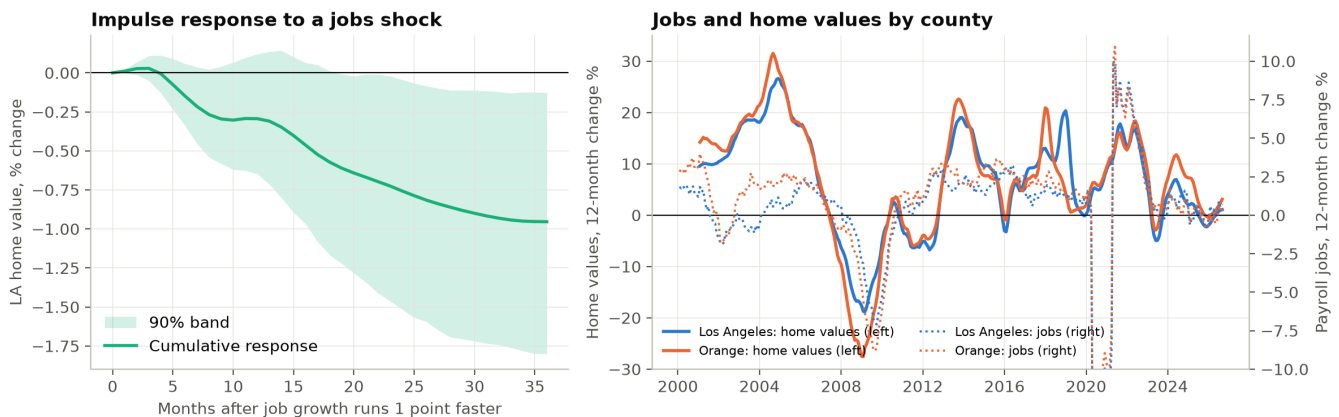
Rates help predict LA home values, but home values do not help predict rates. In the VAR of monthly home-value growth and rate changes, the Granger test of rates on prices gives $F = 1.80$ ($p = 0.05$), and the reverse gives $F = 0.94$ ($p = 0.51$).



The impulse response traces a 1-point rate rise through the VAR, as in Assignment 4 (a unit, non-orthogonalized shock). Values barely move for a few months, then fall steadily: -1.6% after 6 months, -4.5% after 12, -9.2% after 24 and -11.2% after 36. The 90% band stays below zero from month 12 on. This is larger than the ARIMAX estimate because the VAR lets the shock feed through LA's very persistent growth, but both say the same thing: the effect arrives slowly, over one to three years.

Do local jobs drive LA prices? Payroll jobs in the VAR and ARIMAX (slides 6, Assignment 4)

Local jobs don't improve the forecast, and their link to prices rests on the 2020-21 COVID swings. We added 12-month payroll job growth for LA and Orange County to ARIMAX and to the VAR. In the rolling test ARIMAX + jobs placed third (2.81%), behind ARIMAX with rates alone (2.67%). AIC picked a 12-month job lag, and the job coefficient is small, with the wrong sign (-0.016 , $p = 0.29$).



In the VAR, jobs pass the Granger test for prices over 2000–2026, while prices do not predict jobs ($p = 0.70$). The impulse response (left) has the wrong sign, though: faster job growth is followed by slightly lower values. Both results depend on 2020–21, when LA-area payrolls fell 17% and then rebounded 10%, far outside their -7.5% to $+3.1\%$ range in 2000–2019. Lag order is chosen by AIC on each sample:

VAR sample	Jobs → prices (p)	Rates → prices (p)	1-point rate rise, effect after 24 months
2000–2026	0.04	0.05	–9.0%
2000–2019	0.48	0.07	–10.7%
2000–2026 without Mar 2020–Dec 2021	0.08	0.26	–5.3%

Without the COVID months the jobs signal disappears ($p = 0.48$ before 2020). The rate effect survives: a 1-point rise still means about 9% lower values two years later, or 11% in the pre-2020 data. It is weaker only when the 2020–21 months are dropped, which removes the 2021 low-rate boom.

By county, jobs and home values move together in the same year (correlation 0.28 in LA County, 0.44 in Orange County), but job growth does not lead home values (correlations near zero at 12 months). Jobs don't explain the current split either. Over the last year Orange County added fewer jobs than LA County ($+0.3\%$ vs $+0.7\%$), yet its home values rose faster ($+3.1\%$ vs $+1.1\%$).

Who fell hardest in 2022, and is it a stable pattern? Cross-section and panel models (slides 1–2, Assignment 1)

Like the Assignment 1 COVID model, we regressed each neighborhood's 2022–23 drop on its pre-shock price, its county and the size of its 2020–22 boom (496 neighborhoods, robust standard errors, $R^2 = 0.37$). The coefficient on log home value in December 2021 is -1.79 ($t = -7.7$), on Orange County -31.38 ($t = -5.8$), on their interaction $+2.44$ ($t = +6.2$), and on the boom -0.035 ($t = -3.0$).

In LA County, doubling a neighborhood's price meant a 1.2-point deeper drop. In Orange County the slope flips ($-1.79 + 2.44 = +0.65$), so pricier neighborhoods there fell slightly less. Neighborhoods that boomed more also fell more: 10 extra points of boom meant a 0.35-point deeper drop.

Is this a lasting pattern? A panel model of monthly growth for the 405 full-history neighborhoods, with neighborhood and month fixed effects (errors clustered by neighborhood), tests whether pricier neighborhoods are systematically more sensitive to rate changes. The interaction of rate changes and price flips sign across lags of 0 to 24 months and is tiny (at most 0.006 points of monthly growth). Over 26 years there is no stable price-tier effect; the 2022 pattern is specific to that shock.

How much history should we train on? Structural breaks and training windows (slides 1, 3, 8)

The lectures pull in two directions: long data beats short data (slide 1), but structural breaks can make old data misleading (slide 8). We tested both.

Chow tests on an AR(1) model of monthly growth find only a borderline break at the February 2012 trough ($F = 3.06$, $p = 0.05$). The June 2006 peak ($p = 0.22$), March 2020 ($p = 0.68$) and April 2022 ($p = 0.14$) show none, and the largest break statistic across all dates is modest ($F = 6.6$, July 2008). The dynamics of LA's growth have been fairly stable, so old data still carries information.

The rolling test confirms it. Training ARIMA on all data since 2000 gave the lowest error (2.77%), ahead of starting in 2016 (2.97%), 2012 (3.19%) or 2008 (3.31%). Data that includes a full boom and bust teaches the model how fast LA swings, which shorter windows cannot.

Takeaways

1. **For a monthly housing index, simple beats complex.** ARIMA-family models won the fair test. LSTM, GRU, XGBoost, LightGBM and the Chronos foundation model never beat them, and a naive drift was just as good a year out.
2. **How you test decides who wins.** One blind split crowned LightGBM. Eighteen rolling origins ranked it tenth. A comparison built on a single test window, or on 1-month errors of a smoothed index, can make almost any model look good.
3. **An LLM can't be backtested on years it has read about.** Claude, GPT and Gemini beat every model when they could recognize the period, and fell back to ARIMAX's level or worse when they couldn't. Their real skill is only measurable on dates after their training cutoff.
4. **Mortgage rates matter, slowly.** A 1-point rise lowers LA values about 9% over two years, and rates lead prices rather than the reverse. The effect is too slow to help much beyond the 12-month horizon, and local payroll jobs add nothing once rates are in the

model.

5. **Use the long history.** Training on all data since 2000 forecast better than any shorter window, and break tests find LA's growth dynamics fairly stable.

6. **The 2022 shock split LA by price and place, but that split is not permanent.**

Expensive LA County neighborhoods fell hardest in 2023, yet the high end led again over the last year, and 26 years of panel data show no stable price-tier effect. Each neighborhood needs its own model.

For a buyer or lender, the practical message is: expect about +5% for the typical LA home over the next year, but with a range from -6% to +16%. Look at the specific neighborhood, not the metro number.

Limitations and next steps

- **Smoothed data.** Zillow's smoothed, seasonally adjusted index hides month-to-month noise and seasonality. Raw ZHVI or the Case-Shiller index would give a harder test.
- **One overlapping test period.** The 18 rolling origins share data, so their errors are correlated. A Diebold-Mariano test would show whether ARIMAX's 0.1-point lead over ARIMA is statistically meaningful. We expect it is not.
- **Few explanatory variables.** We used mortgage rates and payroll jobs. Income, housing permits, inventory and rents (Zillow ZORI) are natural additions to the ARIMAX or VAR.
- **One run per LLM prompt.** LLM answers vary from run to run, and Gemini 3.1 Pro was still a preview version. Averaging several runs would steady the scores. The clean test is forward: ask each LLM for September 2026 to August 2027 now, and score it next year.
- **Neural models were lightly tuned.** One architecture, no hyperparameter search. Better tuning might narrow the gap, though the data size limits them.
- **The factor model used two factors and no neighborhood-specific dynamics.** A version that adds a separate AR term per neighborhood on top of the shared factors would keep the local trends we lost.
- **Neighborhood forecasts ignore one-off events** such as the 2025 fires and new development.

How we used AI tools

AI tools wrote much of our code and drafted parts of this report. We checked every result they produced, and two of those checks changed our conclusions.

Tool	What it did
Claude Code (Claude Opus 5.5, Anthropic)	Our main coding assistant. It wrote and debugged the Python notebook, ran the models, drew the charts, built the slide deck and the interactive neighborhood map, and drafted parts of this report.
Chronos-Bolt (Amazon)	A pretrained time-series foundation model, used zero-shot as one of our 14 forecasting models.
Claude Opus 5.5, GPT-5.5, Gemini 3.1 Pro	Asked directly for forecasts and scored like any other model. Prompts and raw answers are in the repository.

We reviewed the code, reran the notebook end to end, and checked every number in this report and the slides against its output. AI output was not taken at face value. The LLMs' first backtest beat every model we built; we traced that to memorized history and redesigned the test. The first jobs result said jobs Granger-cause prices; a robustness check showed it rested on the 2020–21 COVID swings.

Appendix: reproducibility

Everything runs from one Python notebook, top to bottom, in about 10 minutes on a laptop. The LLM forecasts are collected separately (they need API keys) and saved as text, so the notebook only scores them.

The repository holds the notebook (`la_home_value_forecasting.ipynb` , plus the same code as a `.py` script), the data in `data/` (Zillow metro and LA-neighborhood ZHVI, neighborhood map positions, the FRED mortgage rate and BLS payroll jobs), the LLM scripts (`llm_prompts.py` , `llm_run_api.py`) with every prompt and raw answer in `llm/` , all charts and result tables in `figures/` , and package versions in `requirements.txt` . The full neighborhood-level ZHVI file is 103 MB, so only the LA subset is included; the full file is on [Zillow Research](#) (ZHVI, all homes, mid tier, smoothed and seasonally adjusted, by neighborhood).

Sources: [Zillow Home Value Index](#) · [FRED, 30-Year Fixed Rate Mortgage Average \(MORTGAGE30US\)](#) · Hyndman and Athanasopoulos, *Forecasting: Principles and Practice* · Stock and Watson (2002), "Forecasting Using Principal Components From a Large Number of Predictors," *JASA* 97(460).